

PAPER_710: NGC 2014 + NGC 2020: Tapestry of Blazing Starbirth UQFF

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Abstract

NGC 2014 (red, HII emission) and NGC 2020 (blue, WR-dominated) form the iconic Tapestry of Blazing Starbirth in the Large Magellanic Cloud (163,000 ly). They represent a dual-arm star-forming complex driven by Wolf-Rayet and OB stellar winds with rapid secondary star formation. The UQFF MUGE is derived for this system.

1. System Parameters

Parameter	Value	Description
M_0	$240M_\odot$	Initial OB/WR stellar mass
r_{region}	9.461×10^{16} m	Region span (10 ly)
$\dot{M}_{SF}$	41.67	Secondary SF mass fraction
τ_{SF}	1.578×10^{14} s	SF timescale (5 Myr)
B_{LMC}	10^{-6} T	LMC magnetic field

2. Master UQFF Gravity Equation

$$g_{TB}(t) = \frac{GM(t)}{r^2}(1+H_0t) \left(1 - \frac{B}{B_{crit}}\right) + (U_{g1}+U_{g4})(1+f_{TRZ}) + \frac{\Lambda c^2}{3} + q(v \times B) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-10}$$

2.1 Secondary Star Formation (WR + OB triggered)

$$M(t) = M_0(1 + 41.67 e^{-t/\tau_{SF}})$$

At $t = 2.5$ Myr: $M \approx 240(1 + 41.67 \cdot 0.709) \approx 240 \times 30.7 \approx 7368M_\odot$

3. Numerical Result

$$g_{TB}(t = 2.5 \text{ Myr}) \approx 1.053 \times 10^{-4} \text{ m/s}^2$$

References

- Hubble WFC3 image (2020)
- Crowther (2007), ARA&A, 45, 177 (Wolf-Rayet stellar physics)
- UQFF Framework, Star-Magic Session 175

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